Financial Performance Measures

Notation: TA: total assets; CA: current assets; CL: current liabilities; E: equity; LTD: long-term debt; NI: net income; $ATOI$: after-tax operating income; $COGS$: cost of goods sold.

Performance measures	
Market value added	Market value of equity − book value of equity.
Market-to-book*	Market value of equity / book value of equity.
Profitability measures	
Return on assets (ROA)*	$ATOI/TA$.
Return on capital (ROC)	$ATOI/(LTD+E)$.
Return on equity (ROE)*	NI/E.
Economic value added (EVA)	$ATOI$ − (Cost of capital × Capital).
Efficiency measures	
Asset turnover	Sales / total assets at start of year.
Receivables turnover	Sales / receivables at start of year.
Average collection period	Receivables at start of year / daily sales.
Inventory turnover	COGS / inventory at start of year.
Days in inventory	Inventory at start of year / daily COGS.
Profit margin	NI/Sales.
Operating profit margin	$ATOI$/Sales.
Leverage measures	
Long-term debt ratio	$LTD/(LTD+E)$.
Long-term debt–equity	LTD/E.
Total debt ratio	Total liabilities / total assets.
Times interest earned	EBIT / interest payments.
Cash coverage ratio	$(EBIT$ + Depreciation$)$/Interest.
Liquidity measures	
NWC to assets	NWC/TA.
Current ratio	CA/CL.
Cash ratio	(Cash + marketable securities) / CL.
Quick ratio	(Cash + marketable securities + receivables) / CL.

* Alternative formulas below / in the next column. Daily sales = annual sales / 365; daily COGS = $COGS/365$. EVA uses monetary units ($ millions in the source).

Market and Book Value

$$\text{Market cap} = \text{Number of shares} \times \text{Market price per share}$$

$$\text{Market-to-book} = \frac{\text{Market cap}}{\text{Book equity}} = \frac{\text{Market price per share}}{\text{Book value per share}}$$

$$\text{Book value per share} = \frac{\text{Book equity}}{\text{Number of shares}}$$

Book value: accounting / balance-sheet value; backward looking. **Market value:** current value investors place on equity; forward-looking resale value. The source's equity-based "price per share" formula is **book value per share**.

Core Equations and Cash Flow

Further notation: I: interest; D: depreciation; t: tax rate; CFO: cash flow from operations; CFI: cash flow from investments.

Income and Tax

$$EBIT = \text{Revenue} + \text{Other income} - \text{Cost} - D$$

Do **not** subtract taxes or interest. Net sales is the revenue line.

$$ATOI = EBIT(1-t) = NI + I(1-t)$$

$$NI = EBIT - I - \text{Taxes} \qquad t = \frac{\text{Taxes}}{EBIT - I}$$

Dividends do **not** reduce net income and are not expenses.

Assets, Capital and Working Capital

$$TA = \text{Total liabilities} + E \qquad E = TA - \text{Total liabilities}$$

$$NWC = CA - CL \qquad \text{Capital} = \text{Debt} + E$$

$$LTD = \text{Long-term debt ratio} \times (TA - CL)$$

Use **total assets**, not current assets, in $TA - CL = LTD + E$.

ROA and ROE Decompositions*

$$ROA = \text{Asset turnover} \times \text{Operating margin} = \frac{ATOI}{TA}$$

$$ROE = \underbrace{\frac{TA}{E}}_{\text{Leverage}} \times \underbrace{\frac{\text{Sales}}{TA}}_{\text{Asset turnover}} \times \underbrace{\frac{ATOI}{\text{Sales}}}_{\text{Operating margin}} \times \underbrace{\frac{NI}{ATOI}}_{\text{Debt burden}}$$

Use consistent asset bases; the ROE product simplifies to NI/E.

Days in Inventory

$$\text{Days in inventory} = \frac{\text{Inventory}}{COGS/365} = \frac{365}{\text{Inventory turnover}}$$

Operating and Free Cash Flow

$$CFO = NI + D - \Delta NWC$$

$$FCF = NI + I + D - \Delta NWC + CFI = I + CFI + CFO$$

FCF follows the source's convention. Investment spending is negative CFI; CapEx is a cash outflow. Operating changes in NWC exclude **cash and short-term debt**.

Increase in	FCF	Increase in	FCF
Receivables	↓	Depreciation	↑*
Inventory	↓	CapEx	↓
Payables	↑		

AR: they owe us → cash tied up. **AP:** we owe them → cash retained. $\uparrow NWC \rightarrow \downarrow$ Cash flow. * Depreciation is non-cash; its cash benefit is a tax shield when it reduces tax.

Earnings, Dividends and Shares

$$\Delta\text{Retained earnings} = NI - \text{Dividends} \qquad EPS = \frac{NI}{\text{Shares}}$$

No stock issued: the source attributes the equity increase to retained earnings. If $\Delta E = \$210$ and dividends $= \$100$, then $NI = \$310$. Use **net income**, not retained earnings, for EPS.

Sample Statements ($)

Balance Sheet

Assets	$	Liabilities / equity	$
Current assets		**Current liabilities**	
Cash	17	Debt due for repayment	27
Receivables	45	Payables	37
Inventories	52	**Total current liabilities**	**64**
Total current assets	**114**	Long-term debt	360
Fixed assets		**Total liabilities**	**424**
Property, plant & equipment	530	**Shareholders' equity**	**98**
Less accum. depreciation	122		
Net tangible fixed assets	**408**		
Total assets	**522**	**Total liabilities & equity**	**522**

Income Statement

Line item	Amount
Net sales	720
Cost of goods sold	590
Selling, general & administrative expense	40
Depreciation	14
EBIT (earnings before interest and taxes)	**76**
Interest expense	27
Taxable income	**49**
Federal plus other taxes	17
Net income	**32**

Corporate Finance Theory

Ownership and Liability

A **corporation** is a separate legal entity. **Shareholders** own it; **managers** run it; the **board of directors** monitors management for shareholders. Shareholders have **limited liability** (no personal liability for corporate obligations). Corporations have an **unlimited lifespan**: share transfers do not end the business. A **sole proprietor** owns and controls the business alone; a **partnership** has at least two people. Both have **unlimited liability** in the course's framework.

Investment vs Financing

Investment: which real assets to buy. **Financing:** how to fund them. **Capital structure:** mix of long-term debt and equity. **Real assets** produce goods / services; **financial assets** (stocks / bonds) claim income from real assets. Buying another company is an investment decision; issuing shares to pay for it is financing. One transaction can involve both.

Agency and Governance

Managers are shareholders' **agents** but may pursue their own interests. **Agency costs:** value lost from these conflicts plus mitigation costs. Least likely in a sole proprietorship (owner = manager). **Executive incentives** and **corporate governance** help control conflicts: board oversight, legal requirements, activist shareholders, takeovers, and accounting / reporting standards. **Goal: maximise current shareholder wealth / market value**, not simply accounting profit.

Accounting, Cash Flow and Tax

Accrual accounting: profit ≠ cash flow; record transactions when earned / incurred, not when cash changes hands. Cash flows are **operating, investing, or financing**. **Marginal tax rate:** tax on the next dollar earned. **Average tax rate:** total tax / income.